# Type MS · Describe phagocytosis, antibody production and barriers to pathogens

## Answer 1 of 9 · Question 6 part a.ii · 3 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

| Question Number | | | | | |
|---|---|---|---|---|---|
| 6 | (a) | (i) | phagocytosis; | | 1 |
| | | (ii) | • enclosed bacterium fuses with lysosome;<br>• enzymes break down /digests bacterium;<br>• remains of bacterium removed from white blood cell; | | 3 |
| | | (iii) | • label/identify/complementary to bacterium/antigens presented;<br>• so response quicker if new infection/better immune response; | | 2 |
| | (b) | (i) | any three from<br>• prick skin to obtain blood;<br>• use a sterile needle/pin<br>• smear/spread blood on slide;<br>• ensure smear is thin;<br>• add stain;<br>• allow to dry; | | 3 |
| | | (ii) | any four from<br>• red blood cells would be seen as circle/not in 3D;<br>• white blood cells would be larger than red/rbc too large;<br>• wouldn't see platelets;<br>• because too small for light microscope; | | 4 |
| | | (iii) | • no risk of infection;<br>• e.g. AIDS; | R contaminate | 2 |

Total 15

## Type MS · Describe phagocytosis, antibody production and barriers to pathogens

### Answer 1 of 9 · Question 6 part a.ii · 3 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 2 of 9 · Question 2 part a.i · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) (i) |  | | 1 |
| (ii) | In the following order only:<br><br>phagocytes;<br>enzymes; | | <br><br>1<br>1 |
| (iii) | produce antibodies/release antitoxins; | | 1 |
| (b) | <table><tr><th>Bacterial cell</th><th>Human skin cell</th></tr><tr><td></td><td>✓</td></tr><tr><td>✓</td><td>✓</td></tr><tr><td></td><td></td></tr><tr><td>✓</td><td></td></tr></table> | One mark for each correct row | 3 |
| (c) | **D** (a single-stranded helix containing the bases AUGC);<br><br>A is incorrect as RNA is not a double-stranded helix<br>B is incorrect as RNA is not a double-stranded helix<br>C is incorrect as RNA does not contain the base T | | 1 |

Total question 2 = 8 marks

## Type MS · Describe phagocytosis, antibody production and barriers to pathogens

### Answer 2 of 9 · Question 2 part a.i · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 3 of 9 · Question 2 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) (i) |  | | 1 |
| (ii) | In the following order only:<br><br>phagocytes;<br>enzymes; | | <br><br>1<br>1 |
| (iii) | produce antibodies/release antitoxins; | | 1 |
| (b) | <table><tr><th>Bacterial cell</th><th>Human skin cell</th></tr><tr><td></td><td>✓</td></tr><tr><td>✓</td><td>✓</td></tr><tr><td></td><td></td></tr><tr><td>✓</td><td></td></tr></table> | One mark for each correct row | 3 |
| (c) | **D** (a single-stranded helix containing the bases AUGC);<br><br>A is incorrect as RNA is not a double-stranded helix<br>B is incorrect as RNA is not a double-stranded helix<br>C is incorrect as RNA does not contain the base T | | 1 |

Total question 2 = 8 marks

## Type MS · Describe phagocytosis, antibody production and barriers to pathogens

### Answer 3 of 9 · Question 2 part a.ii · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 4 of 9 · Question 2 part a.iii · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) (i) |  | | 1 |
| (ii) | In the following order only:<br><br>phagocytes;<br>enzymes; | | <br><br>1<br>1 |
| (iii) | produce antibodies/release antitoxins; | | 1 |
| (b) | <table><tr><th>Bacterial cell</th><th>Human skin cell</th></tr><tr><td></td><td>✓</td></tr><tr><td>✓</td><td>✓</td></tr><tr><td></td><td></td></tr><tr><td>✓</td><td></td></tr></table> | One mark for each correct row | 3 |
| (c) | **D** (a single-stranded helix containing the bases AUGC);<br><br>A is incorrect as RNA is not a double-stranded helix<br>B is incorrect as RNA is not a double-stranded helix<br>C is incorrect as RNA does not contain the base T | | 1 |

Total question 2 = 8 marks

## Type MS · Describe phagocytosis, antibody production and barriers to pathogens

### Answer 4 of 9 · Question 2 part a.iii · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Describe phagocytosis, antibody production and barriers to pathogens

### Answer 5 of 9 · Question 6 part a.i · 1 mark

2023 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 6 | (a) | (i) | phagocytosis; | | 1 |
| | | (ii) | • enclosed bacterium fuses with lysosome;<br>• enzymes break down /digests bacterium;<br>• remains of bacterium removed from white blood cell; | | 3 |
| | | (iii) | • label/identify/complementary to bacterium/antigens presented;<br>• so response quicker if new infection/better immune response; | | 2 |
| | (b) | (i) | any three from<br>• prick skin to obtain blood;<br>• use a sterile needle/pin<br>• smear/spread blood on slide;<br>• ensure smear is thin;<br>• add stain;<br>• allow to dry; | | 3 |
| | | (ii) | any four from<br>• red blood cells would be seen as circle/not in 3D;<br>• white blood cells would be larger than red/rbc too large;<br>• wouldn't see platelets;<br>• because too small for light microscope; | | 4 |
| | | (iii) | • no risk of infection;<br>• e.g. AIDS; | R contaminate | 2 |

Total 15

## Type MS · Describe phagocytosis, antibody production and barriers to pathogens

### Answer 5 of 9 · Question 6 part a.i · 1 mark

2023 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 6 of 9 · Question 8 part a.i · 1 mark

2025 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 8 (a) | (i) | cilia/skin/mucus | allow other suitable answers | 1 |
| | (ii) | kills/inhibits growth of pathogens/damage cell membranes; | | 1 |
| | (iii) | • (phagocytes) engulf pathogens/microorganisms / bacteria;<br>• digest/breakdown pathogens/microorganisms / bacteria; | | 2 |
| | (iv) | antigen/immunogen; | | 1 |
| | (v) | • antibodies have a particular shape /are complementary (to antigen);<br>• only recognise/attach/bind to, one particular/specific pathogen / antigen/receptor; | | 2 |
| | (vi) | • natural passive immunity the body receives antibodies naturally; | accept through breast milk/placenta | |
| | | • artificial passive immunity are when antibodies are injected into the body; | accept e.g. to give immediate protection against a disease; | 2 |
| (b) | (i) | 360 ÷ 30 = 12;<br>108 ÷ 12;<br>9; | | 3 |
| | (ii) | reduce the risk of disease transmission/infection; | | 1 |
| | (iii) | any two from<br>• improper/overuse of antibiotics/over prescribed;<br>• bacterial DNA mutates/bacteria adapt;<br>• some survive and reproduce | | 2 |
| | | | **Total for Question 8 = 15 marks** | |

## Type MS · Describe phagocytosis, antibody production and barriers to pathogens

### Answer 7 of 9 · Question 8 part a.ii · 1 mark

2025 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 8 (a) | (i) | cilia/skin/mucus | allow other suitable answers | 1 |
| | (ii) | kills/inhibits growth of pathogens/damage cell membranes; | | 1 |
| | (iii) | • (phagocytes) engulf pathogens/microorganisms / bacteria;<br>• digest/breakdown pathogens/microorganisms / bacteria; | | 2 |
| | (iv) | antigen/immunogen; | | 1 |
| | (v) | • antibodies have a particular shape /are complementary (to antigen);<br>• only recognise/attach/bind to, one particular/specific pathogen / antigen/receptor; | | 2 |
| | (vi) | • natural passive immunity the body receives antibodies naturally;<br>• artificial passive immunity are when antibodies are injected into the body; | accept through breast milk/placenta<br>accept e.g. to give immediate protection against a disease; | 2 |
| (b) | (i) | 360 ÷ 30 = 12;<br>108 ÷ 12;<br>9; | | 3 |
| | (ii) | reduce the risk of disease transmission/infection; | | 1 |
| | (iii) | any two from<br>• improper/overuse of antibiotics/over prescribed;<br>• bacterial DNA mutates/bacteria adapt;<br>• some survive and reproduce | | 2 |
| | | | **Total for Question 8 = 15 marks** | |

## Type MS · Describe phagocytosis, antibody production and barriers to pathogens

### Answer 8 of 9 · Question 8 part a.iii · 2 marks

2025 JUN · Paper 02 (regional R) · find part a.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 8 (a) | (i) | cilia/skin/mucus | allow other suitable answers | 1 |
| | (ii) | kills/inhibits growth of pathogens/damage cell membranes; | | 1 |
| | (iii) | • (phagocytes) engulf pathogens/microorganisms / bacteria;<br>• digest/breakdown pathogens/microorganisms / bacteria; | | 2 |
| | (iv) | antigen/immunogen; | | 1 |
| | (v) | • antibodies have a particular shape /are complementary (to antigen);<br>• only recognise/attach/bind to, one particular/specific pathogen / antigen/receptor; | | 2 |
| | (vi) | • natural passive immunity the body receives antibodies naturally; | accept through breast milk/placenta | |
| | | • artificial passive immunity are when antibodies are injected into the body; | accept e.g. to give immediate protection against a disease; | 2 |
| (b) | (i) | 360 ÷ 30 = 12;<br>108 ÷ 12;<br>9; | | 3 |
| | (ii) | reduce the risk of disease transmission/infection; | | 1 |
| | (iii) | any two from<br>• improper/overuse of antibiotics/over prescribed;<br>• bacterial DNA mutates/bacteria adapt;<br>• some survive and reproduce | | 2 |
| | | | **Total for Question 8 = 15 marks** | |

Type MS · Describe phagocytosis, antibody production and barriers to pathogens

Answer 9 of 9 · Question 4 part b.ii · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 4 (a) | • diarrhoea by clean water/safe water/ bottled water/boiled water/ sewage treatment/chlorination/good sanitation/not mixing sewage with drinking water/washing hands/good hygiene/ cooking food thoroughly;<br>• malaria by nets/draining swamps/insecticides/ insect repellent's/wearing clothing to cover arms (and legs/mosquito coils/anti-malarial/vaccination ; | Ignore ORT | 2 |
| (b) (i) | antibiotics; | | 1 |
| (ii) | • elderly have weakened/lowered immune system/immunocompromised;<br>• indoor spaces makes transmission easier/reduced air flow/ reduced ventilation/build- up of air droplets with pathogens/ people closer together; | | 2 |
| (c) (i) | • diet;<br>• exercise;<br>• smoking:<br>• drinking alcohol:<br>• stress | Accept drugs | 2 |
| (ii) | • 5 000 000 / 100 000 = 50;<br>• 50 × 445 = 22 250;<br>• 22 300;<br>***22,250 scores 2 marks***<br>***22,300 scores 3 marks*** | ***1 mark for ×445 if there are no other marks*** | 3 |
| (iii) | • use of 445 and 185;<br>• 445 / 185 ;<br>• 2.41:1/2.405:!/2.4:1= 3 marks; | ecf from mp1<br>**if they have done the correct calculation and 2:1 is on the answer line they score** 2 marks | 3 |
| (iv) | • people live longer/elderly population;<br>• these diseases are more common in older people;<br>• less likely to die younger from infectious diseases/better medical facilities/treatment;<br>• different diet/ more meat/saturated fats/processed food;<br>• drinking more alcohol; | People live longer so there is more cancer and dementia/these diseases= 2 marks | 2 |

**Total 15 marks**

## Type MS · Describe phagocytosis, antibody production and barriers to pathogens

### Answer 9 of 9 · Question 4 part b.ii · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |